Tal Boger

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Education

Ph.D., Psychological & Brain Sciences, Johns Hopkins University — 2023 — present
Advisor: Chaz Firestone

BA, Statistics & Data Science, Yale University — 2019 — 2023
with distinction

Publications (* equal contribution)

Journal Articles

Boger, T. (in press). Compression without compromise [commentary]. *Behavioral and Brain Sciences*.

Boger, T., & Firestone, C. (in press). Complexity is a unified cognitive kind. *Nature Human Behaviour*.

Boger, T., & Firestone, C. (2026). Perceiving animacy in 'identical' images. *eLife*, *15*, RP111179.

Boger, T., & Firestone, C. (2025). The psychophysics of style. *Nature Human Behaviour*, *9*(12), 2497-2509.

Boger, T., & Firestone, C. (2025). Visual anagrams reveal high-level effects with 'identical' stimuli. *Current Biology*, *35*(19), R910-R912.

***Boger, T.**, *Yousif, S. R., McDougle, S. D., & Rutledge, R. B. (2025). Random behavior is stable across tasks and time. *Journal of Experimental Psychology: General, 154*(6), 1571-1582.

Boger, T., & Strickland, B. (2025). Object persistence explains event completion. *Cognition*, *259*, 106110.

Boger, T., & Keil, F. C. (2025). Mechanistic complexity is fundamental: Evidence from judgments, attention, and memory. *Journal of Experimental Psychology: General*, *154*(2), 596-606.

Boger, T., & Franconeri, S. L. (2024). Reading a graph is like reading a paragraph. *Journal of Experimental Psychology: General*, *153*(7), 1699-1704.

Li, Y., Wang, Y.Q., **Boger, T.**, Smith, K., Gershman, S.J., & Ullman, T. (2023). An approximate representation of objects underlies physical reasoning. *Journal of Experimental Psychology: General*, *152*(11), 3074-3086.

Boger, T., & Ullman, T. (2023). What is "where": Physical reasoning informs object location. *Open Mind, 7*, 130-140.

*Hafri, A., ***Boger, T.**, & Firestone, C. (2022). Melting ice with your mind: Representational momentum for physical states. *Psychological Science, 33*(5), 725-735.

Refereed Conference Proceedings

Boger, T., Most, S.B., & Franconeri, S.L. (2021). Jurassic mark: Inattentional blindness for a datasaurus reveals that visualizations are explored, not seen. In *2021 IEEE Visualization Conference (VIS)* (pp. 71-75). (**Best Short Paper Award**.)

Boger, T., Ananthabhotla, I., & Paradiso, J. (2021). Manipulating causal uncertainty in sound objects. In *Proceedings of the 16th International Audio Mostly Conference* (pp. 9-15).

Conference Presentations

Boger, T., & Firestone, C. (2026). Perceiving animacy in 'identical' images. Poster presented at the annual meeting of the *Society for Philosophy and Psychology*, 6/18/2026, Baltimore, MD.

Boger, T., & Firestone, C. (2026). Perceiving animacy in otherwise-identical images. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/18/2026, St. Pete Beach, FL.

Firestone, C., & **Boger, T.** (2026). Phrasal momentum. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2026, St. Pete Beach, FL.

Boger, T., Firestone, C., & Liu, S. (2026). Who drew this? The development of visual style perception. Poster presented at the biennial meeting of the *Cognitive Development Society*, 4/10/2026, Montreal, Canada.

Boger, T., Firestone, C., & Liu, S. (2025). Who drew this? Children appreciate visual style differently than adults. Poster presented at the annual meeting of the *Cognitive Science Society*, 8/2/2025, San Francisco, CA.

Boger, T., & Firestone, C. (2025). Complexity is a unified cognitive kind. Talk given at the annual meeting of the *Society for Philosophy and Psychology*, 6/19/2025, Ithaca, NY.

Liu, S., Firestone, C., & **Boger, T.** (2025). Who drew this? Children appreciate visual style differently than adults. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/20/2025, St. Pete Beach, FL.

Boger, T., Liu, S., & Firestone, C. (2025). Complexity is a cognitive universal: Evidence from cross-modal transfer. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2025, St. Pete Beach, FL.

Firestone, C., & **Boger, T.** (2025). Controlling for everything: Canonical size effects with identical stimuli. Talk given at the annual meeting of the *Vision Sciences Society*, 5/17/2025, St. Pete Beach, FL.

Boger, T., & Firestone, C. (2024). The psychophysics of style. Talk given at the annual meeting of the *Object Perception, Attention, and Memory Conference*, 11/21/2024, New York, NY. (**Best Talk Award**.)

Boger, T., & Firestone, C. (2024). The psychophysics of style. Talk given at the annual meeting of the *Society for Philosophy and Psychology*, 6/20/2024, West Lafayette, IN. (**William James Prize**.)

Boger, T., & Firestone, C. (2024). The psychophysics of style. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/18/2024, St. Pete Beach, FL.

Franconeri, S. L., & **Boger, T.** (2024). When processing relationships, visual processing capacity is far less than four. Talk given at the annual meeting of the *Vision Sciences Society*, 5/18/2024, St. Pete Beach, FL.

Hafri, A., **Boger, T.**, & Firestone, C. (2022). Remembering the future: Representational momentum for physical states. Talk given at the annual meeting of the *(European) Society for Philosophy and Psychology*, 7/21/2022, Milan, Italy.
Boger, T., & Keil, F. C. (2022). Mechanism influences intuitive object complexity. Talk given at the annual meeting of the *(European) Society for Philosophy and Psychology*, 7/19/2022, Milan, Italy.

Boger, T., & Firestone, C. (2022). Automatic simulation of unseen physical events. Poster presented at the annual meeting of the *Vision Sciences Society*, 5/17/2022, St. Pete Beach, FL.

Boger, T., Most, S.B., & Franconeri, S.L. (2021). Jurassic mark: Inattentional blindness for a datasaurus reveals that visualizations are explored, not seen. Talk given at the annual (online) meeting of the *IEEE Visualization Conference (VIS)*, 10/26/2021.

Boger, T., Ananthabhotla, I., & Paradiso, J. (2021). Manipulating causal uncertainty in sound objects. Talk given at the annual (online) meeting of the *Audio Mostly Conference*, 9/1/2021.

Boger, T., & Firestone, C. (2021). Through the looking-glass: Visual sensitivity to chirality. Poster presented at the annual (online) meeting of the *Vision Sciences Society*, 5/25/2021.

Hafri, A., **Boger, T.**, & Firestone, C. (2021). Melting ice with your mind: Dynamic representation of physical states. Talk given at the annual (online) meeting of the *Vision Sciences Society*, 5/25/2021.

Boger, T., Hafri, A., & Firestone, C. (2020). Melting ice with your mind: Representational momentum in state-space. Talk given at the annual (online) meeting of the *Object Perception, Attention, and Memory Conference*, 11/19/2020.

Honors & Awards

2026	Walter L. Clark Teaching Award, *Johns Hopkins Psychological & Brain Sciences*
2025	Robert S. Waldrop Jr. Investigator Award, *Johns Hopkins Psychological & Brain Sciences*
2024	Best Talk Award, *Object Perception, Attention, and Memory Conference*
2024	Professional Development Award, *Object Perception, Attention, and Memory Conference*
2024	William James Prize, *Society for Philosophy and Psychology*
2023	Graduate Research Fellowship Program (GRFP), *National Science Foundation*
2022	National Eye Institute Early Career Scientist Travel Grant, *Vision Sciences Society*
2021	Best Short Paper Award, *IEEE Visualization Conference (VIS)*

Invited Talks

2025	Cognitive Tools Lab (Fan lab), *Stanford University*
2024	Developing Minds Lab (Kibbe lab), *Boston University*
2023	Computational Cognitive Neuroscience Lab (Gershman lab), *Harvard University*
2023	Computation and Language Lab (Piantadosi lab), *UC Berkeley*
2022	Vision Sciences Lab (Konkle and Alvarez lab), *Harvard University*
2021	Machine Common Sense Meeting, *DARPA*

Teaching Experience

Instructor	The art of information: Information-theoretic approaches to psychology and art (Intersession, 2026)
Teaching Assistant	Introduction to Psychology (Fall, 2026) Introduction to Psychology (Fall, 2025) Advanced Research Design and Analysis (Spring, 2025) Advanced Statistical Methods (Fall, 2024)
Guest Lecturer	Introduction to Psychology (Fall, 2025; Fall, 2026) Advanced Statistical Methods (Fall, 2024)

Professional Service and Membership

Ad-Hoc Reviewer	Journals: Attention, Perception, & Psychophysics; Cognition; Journal of Experimental Psychology: Human Perception and Performance; Open Mind; Perception; Psychonomic Bulletin & Review; IEEE Transactions on Visualization and Computer Graphics Conferences: Cognitive Science Society; ACM Conference on Human Factors in Computing Systems (SIGCHI)
Societies	Cognitive Development Society; Cognitive Science Society; Society for Philosophy and Psychology; Vision Sciences Society